TahiConnect: An AI-Powered Hybrid Web and Mobile Tailoring Service Management System with Virtual Try-On and Real-Time Order Tracking

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CCS Concepts

CCS → Information Systems → Information Systems Applications → Enterprise Information Systems

CCS → Computing Methodologies → Artificial Intelligence

CCS → Human-Centered Computing → Ubiquitous and Mobile Computing Systems and Tools

General Terms

Management, Design, Human Factors, Performance, Reliability

Keywords

Tailoring Service Management, Virtual Try-On, Real-Time Order Tracking, Artificial Intelligence, Hybrid Web and Mobile System, Measurement Validation

1. INTRODUCTION

1.1 Project Context

The rapid advancement of information technology has transformed how businesses operate, communicate, and deliver services across various industries worldwide [1]. Recent studies show that digital transformation has become a key driver of organizational efficiency by enabling the automation and optimization of business processes [1]. In addition, emerging technologies such as artificial intelligence (AI) and web-based systems enhance customer engagement, support service innovation, and improve overall business competitiveness, particularly among small and medium-sized enterprises (SMEs) [2]. In the Philippines, the adoption of digital technologies among micro, small, and medium enterprises (MSMEs) continues to be encouraged to improve competitiveness, productivity, and service delivery in an increasingly digital economy [3]. Likewise, studies and initiatives within the Davao Region emphasize the importance of technology adoption and digital innovation among local enterprises to support business growth and modernization [4][5]. These developments create opportunities for traditional service-based industries, such as tailoring businesses, to adopt information systems that improve operational efficiency and customer engagement.

Based on interviews and observations conducted with tailoring service providers, most tailoring businesses still rely on manual processes for managing customer information, body measurements, appointment schedules, service orders, and payment records. Customer measurements are commonly recorded on paper forms, while appointment schedules and order updates are communicated through phone calls, text messages, or in-person visits. Tailors and shop personnel manually monitor garment production progress and customer transactions, making the overall process time-consuming and prone to human error.

Several operational challenges were identified during the data-gathering phase. Customer records and body measurements may become misplaced or incorrectly recorded, resulting in fitting errors and production delays. Manual appointment scheduling may lead to scheduling conflicts and missed appointments, while customers have limited visibility regarding the status of their tailoring requests because updates often require direct communication with the tailoring shop. Furthermore, based on interviews and observations conducted with tailoring business owners and staff, existing tailoring-related solutions used by respondents do not provide a unified platform that combines customer profile management, measurement storage, appointment scheduling, AI-powered virtual try-on, measurement validation, payment management, and real-time order tracking. These identified gaps serve as the basis for the proposed system modules and specific objectives of the study.

Table 1. Feature Comparison Matrix

Features	Tailornova	Virtusize	Fit Analytics	TahiConnect (Proposed System)
Customer Profile Management	Partial	Partial	Partial	✓
Measurement Storage	Partial	✓	✓	✓
Appointment Scheduling	✗	✗	✗	✓
Service Order Management	✗	✗	✗	✓
Payment Recording	✗	✗	✗	✓
Real-Time Order Tracking	✗	✗	✗	✓
AI Virtual Try-On	✓	Partial	✗	✓
Measurement Validation	Partial	✓	✓	✓
Mobile Accessibility	Partial	✓	✓	✓
Tailoring Workflow Management	✗	✗	✗	✓

✓ – Fully Supported **Partial** – Feature is available but limited ✗ – Feature is not available

The development of TahiConnect is motivated by the need to address the operational inefficiencies experienced by tailoring businesses in managing customer records, body measurements, appointments, service orders, and production updates. The proposed system is expected to benefit tailoring service providers by improving record accuracy, streamlining business processes, reducing manual errors, and enhancing customer satisfaction through digital services [1]. Furthermore, the study aligns with Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) [8] and supports the University of Mindanao's research thrust on ICT-enabled solutions that promote innovation, organizational efficiency, and digital transformation [5].

1.2. Purpose and Description

This study aims to develop TahiConnect: An AI-Powered Hybrid Web and Mobile Tailoring Service Management System with Virtual Try-On and Real-Time Order Tracking to improve the efficiency of tailoring operations and customer service processes [1][2]. The proposed system supports Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) [8] by promoting the adoption of digital technologies and artificial intelligence within the tailoring industry. The study also aligns with the University of Mindanao's research agenda on ICT-enabled systems that improve organizational productivity and service delivery [5].

TahiConnect is a hybrid web and mobile information system designed for tailoring service providers and customers in Davao City. The system consists of three user roles: **Administrator, Tailoring Staff, and Customer**. The Administrator has full access to manage customer records, appointments, service orders, payments, reports, and system settings. Tailoring Staff are responsible for updating measurements, processing garment requests, managing production progress, and monitoring customer transactions. Customers can create accounts, manage their profiles, store body measurements, upload garment design references, schedule appointments, track orders, view payment records, and utilize the AI-powered virtual try-on feature. Through these functionalities, the system provides a centralized platform that improves communication, operational efficiency, and customer satisfaction.

1.3. Objectives

1.3.1 General Objectives

This study aims to develop and evaluate TahiConnect, an AI-Powered Hybrid Web and Mobile Tailoring Service Management System with Virtual Try-On and Real-Time Order Tracking for tailoring service providers and customers in Davao City.

1.3.2 Specific Objectives

To achieve the general objective, the researchers will specifically develop, implement, and evaluate the following system components:

1.3.2.1. Develop a customer profile and measurement management module using the Laravel Framework and MySQL Database to securely store, retrieve, and manage customer information, body measurements, tailoring preferences, and garment history.

1.3.2.2. Implement an AI-powered virtual try-on module that generates garment previews using customer-uploaded images and stored measurement data to support garment visualization before production.

1.3.2.3. Develop an appointment scheduling and fitting management module using the Laravel Framework that enables customers and tailoring staff to create, manage, and monitor fitting schedules and appointments.

1.3.2.4. Develop a service order and payment management module using the Laravel Framework and MySQL Database that allows customers to submit garment requests, upload design references, manage transactions, and generate payment records.

1.3.2.5. Implement a real-time order tracking and measurement validation module that enables customers to monitor production progress while automatically checking measurement consistency to reduce fitting errors.

1.4. Scope and Limitations

This study focuses on the design, development, and evaluation of TahiConnect, an AI-Powered Hybrid Web and Mobile Tailoring Service Management System intended for tailoring service providers and customers in Davao City. The system covers customer profile management, body measurement recording, garment history management, appointment scheduling, service order processing, payment recording, AI-powered virtual try-on, measurement validation, and real-time order tracking. The system accommodates three primary user roles: Administrator, Tailoring Staff, and Customer. The proposed system will be implemented and evaluated within selected tailoring businesses in Davao City and will manage tailoring-related records and transactions generated during the study period.

The study does not cover advanced garment manufacturing automation, embroidery machine integration, inventory management of fabrics and materials, or highly specialized pattern-making operations. The virtual try-on feature is intended only for garment visualization and does not guarantee exact real-world fitting results, fabric behavior, or garment draping. The reliability of the measurement validation component depends on the accuracy of the customer measurements entered into the system. The evaluation of the system will be limited to selected respondents and participating tailoring businesses involved in the study.

2. METHODS

This chapter presents the methodology used in the development of TahiConnect. The study adopts the Agile Software Development Methodology as the framework for planning, designing, developing, testing, and evaluating the proposed system. Agile promotes iterative development, continuous stakeholder feedback, and flexibility in responding to changing requirements, making it suitable for developing information systems intended to address the dynamic needs of tailoring service providers and customers [6]. Agile methodologies have been widely recognized for improving software quality, stakeholder involvement, and project adaptability in information system development projects [6].

Figure 1. Methods model

The Agile methodology consists of the following phases: Requirements Gathering, Planning, System Design, Development, Testing, Deployment, and Evaluation/Review. Each phase allows the researchers to continuously improve system functionalities based on user feedback and testing results. This iterative approach ensures that the developed system effectively addresses the identified needs and operational challenges of tailoring businesses.

2.1 Data Gathering

2.1.1 Research Participants / Respondents

The respondents of the study consist of tailoring shop owners, tailoring staff, and customers from selected tailoring businesses in Davao City. These participants were selected because they are directly involved in tailoring operations and possess firsthand knowledge regarding customer management, appointment scheduling, body measurement recording, service order processing, and order monitoring activities.

Tailoring shop owners and staff served as key informants during the interview process to provide insights regarding existing operational procedures, challenges, and system requirements. Customers participated in surveys and interviews to identify their experiences, expectations, and preferences regarding tailoring services and digital service platforms.

2.1.2 Research Instruments

The researchers utilized interview guides, survey questionnaires, and observation checklists as primary data-gathering instruments [9]. The interview guide was used to collect qualitative information regarding current tailoring business operations, existing challenges, and desired system features. Survey questionnaires were administered to customers and tailoring personnel to gather quantitative data regarding user needs, satisfaction levels, and technology acceptance.

Observation checklists were also utilized to document existing workflows, record management practices, appointment scheduling procedures, and order tracking activities within participating tailoring businesses. These instruments enabled the researchers to gather comprehensive information necessary for defining system requirements and identifying operational gaps.

2.1.3 Data Gathering Procedure

The researchers first secured permission from participating tailoring businesses before conducting interviews, surveys, and observations. Initial interviews were conducted with tailoring shop owners and staff to understand existing business processes and identify operational challenges. Customer surveys were subsequently administered to gather information regarding service experiences, preferences, and expectations for a digital tailoring service platform.

The collected data were analyzed and consolidated to identify recurring issues, functional requirements, and opportunities for process improvement [9]. The findings served as the basis for developing the proposed system modules, conceptual framework, and technical specifications of TahiConnect.

2.2 Analysis

The analysis phase focused on identifying the system requirements necessary to address the operational challenges identified during data gathering. Information obtained from interviews, surveys, and observations was analyzed to determine the functional and non-functional requirements of the proposed system [10].

2.2.1 User Requirements

Based on the data-gathering activities conducted through interviews, surveys, and observations, several user requirements were identified for the proposed TahiConnect system. For the **Administrator**, the system must provide functionalities for managing customer records and user accounts, overseeing appointments and service orders, monitoring payments, generating reports, and configuring system settings and user permissions. The

Tailoring Staff require features that allow them to record and update customer measurements, process garment requests and production activities, update order statuses and fitting schedules, and monitor assigned customer transactions. Meanwhile, the **Customers** should be able to register and manage their personal accounts, store and update body measurements, schedule appointments and fitting sessions, upload garment design references, track order progress in real time, utilize the AI-powered virtual try-on feature, and view payment records and transaction histories. These requirements serve as the foundation for the design and development of the proposed system modules.

2.2.2 Hardware and Software Requirements

Hardware Requirements

The development and deployment of TahiConnect require specific hardware and software resources to ensure efficient system operation and performance. For the **development environment**, the system requires a computer equipped with at least an Intel Core i5 processor or higher, a minimum of 8GB RAM, 256GB SSD storage, and a stable internet connection to support coding, testing, and database management activities. For the **deployment environment**, the system will be accessible through a web server and can be used on Android smartphones, personal computers, or laptops with internet connectivity. On the software side, the development process utilizes Visual Studio Code as the integrated development environment, Laravel Framework for web application development, MySQL Database Management System for data storage, XAMPP as the local server environment, GitHub for version control, and Figma for user interface design and prototyping. End-users can access the system using common web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge through devices running the Android operating system or other compatible platforms with internet access.

2.2.3 Functional and Non-Functional Requirements

The proposed system includes several functional requirements that define the core services and operations of TahiConnect [11]. These functionalities include user registration and authentication, customer profile management, measurement management, appointment scheduling, service order management, payment recording and monitoring, AI-powered virtual try-on, measurement validation, real-time order tracking, and report generation. These functions are designed to support both business operations and customer interactions within a centralized tailoring service management platform.

In addition to the functional requirements, the system must satisfy several non-functional requirements to ensure quality and reliability. The system should provide a user-friendly interface to enhance usability and user experience [11]. It must maintain accurate records and transaction data to ensure reliability and consistency in system operations [11]. Security mechanisms such as authentication and authorization should be implemented to protect user information and prevent unauthorized access [11]. The system should efficiently process user requests to maintain acceptable performance levels and should remain accessible through web and mobile platforms with internet connectivity to ensure availability [11]. Furthermore, the system should be designed with maintainability in mind, allowing future enhancements, updates, and modifications without significantly affecting existing functionalities [11].

2.2.4 Feasibility Study

Technical Feasibility. The proposed system is technically feasible because the required development tools, frameworks, hardware resources, and internet infrastructure are readily available. Laravel Framework, MySQL Database, and modern web technologies provide sufficient capabilities for implementing the proposed functionalities [6].

Operational Feasibility. The system is operationally feasible because tailoring businesses currently perform these activities manually. The proposed system automates existing processes without significantly changing operational workflows, making adoption easier for users while supporting the digital transformation of service-oriented SMEs [1][2].

Economic Feasibility. The proposed system utilizes open-source technologies such as Laravel, MySQL, XAMPP, and GitHub, reducing software acquisition costs. The anticipated benefits, including improved efficiency, reduced errors, and enhanced customer satisfaction, outweigh the implementation costs.

Schedule Feasibility. The project can be completed within the allocated capstone development timeline through the use of Agile Software Development practices that support incremental development and continuous testing.

2.3 Design

The design phase focuses on translating the identified user requirements into a structured system architecture and functional design for TahiConnect. This phase defines the conceptual framework, system components, database structure, architecture, and user interface prototypes that will guide the development of the proposed system. The design ensures that the system addresses the identified operational challenges while providing an efficient and user-friendly platform for tailoring service providers and customers.

2.3.1 Conceptual Framework

The study utilizes the Input-Process-Output (IPO) Model as its conceptual framework [13]. The IPO Model illustrates how data and resources are transformed through system processes to generate meaningful outputs that support decision-making and operational efficiency [13].

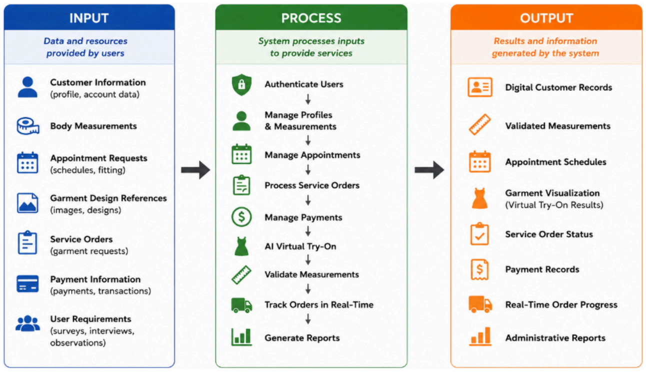

Figure 2. Conceptual Framework

Input

The inputs of the proposed system consist of customer information, body measurements, appointment requests, garment design references, service orders, payment information, user requirements, and data gathered from surveys and interviews. Customer information includes personal details and account credentials used for profile management, while body measurements serve as the basis for tailoring customization and fit validation. Appointment requests, garment design references, service orders, and payment information are provided by customers and tailoring staff throughout the transaction process. Additionally, user requirements and findings from surveys and interviews are utilized to identify system functionalities and ensure that the proposed system addresses the needs and challenges experienced by its intended users.

Process

The system processes involve user authentication and authorization, customer profile management, measurement recording and validation, appointment scheduling, service order processing, payment management, AI-powered virtual try-on processing, real-time order tracking, and report generation. Upon accessing the system, users are authenticated according to their assigned roles and permissions. Customer information and body measurements are recorded and validated to ensure accuracy before proceeding with garment requests. The system then facilitates appointment scheduling, service order processing, and payment management while generating virtual garment previews through the AI-powered virtual try-on feature. Throughout the production process, order status updates are monitored and displayed through the real-time order tracking module. Administrative and operational reports are also generated to support decision-making and business management activities.

Output

The outputs generated by the system include digital customer records, validated body measurements, appointment schedules, garment visualization results, service order status updates, payment records, real-time order progress information, and administrative reports. These outputs provide tailoring businesses with organized and accessible information that supports efficient operations and accurate record management. Customers benefit from improved access to appointment details, garment previews, payment records, and order tracking information, while administrators and tailoring staff can utilize system-generated reports for monitoring business performance and managing daily operations.

2.3.2 Capstone Element

AI-Powered Virtual Try-On with Measurement Validation

The capstone element of TahiConnect is the integration of an AI-Powered Virtual Try-On System with Measurement Validation and Real-Time Order Tracking. This feature distinguishes the proposed system from conventional tailoring management systems by allowing customers to visualize garments digitally before production while validating measurements to reduce fitting errors [7][12].

Most existing tailoring management systems focus primarily on record management, scheduling, and transaction processing. While some fashion technology platforms provide virtual try-on capabilities, they often lack tailoring-specific measurement validation and order management functions. TahiConnect combines these functionalities into a single platform specifically designed for tailoring businesses.

Significance of the Capstone Element

The AI-powered virtual try-on module enables customers to preview garment designs using uploaded images and measurement data before production begins. Studies have shown that virtual try-on technologies improve garment visualization, support sizing accuracy, and help users make more informed clothing decisions before purchase or production [7][12]. This functionality can improve customer confidence in design selections and reduce misunderstandings between customers and tailoring staff.

The measurement validation component evaluates customer measurements against predefined garment measurement requirements. This process helps identify potential fitting issues before garment production, reducing material waste, production revisions, and customer dissatisfaction [7].

Measurement Validation Algorithm

The system evaluates measurement consistency using tolerance-based validation.

Formula:

$$MV = |CM - GM|$$

Where:

- MV = Measurement Variance
- CM = Customer Measurement
- GM = Garment Measurement Standard

Validation Rule:

$$MV \leq T$$

Where:

T = Acceptable Tolerance Value

If the computed variance is less than or equal to the acceptable tolerance value, the measurement is considered valid.

Sample Computation

Customer Chest Measurement: 38 inches

Required Garment Chest Measurement: 39 inches

Tolerance Value: 2 inches

Computation:

$$MV = |38 - 39|$$

$$MV = 1$$

Since:

$$1 \leq 2$$

The measurement passes validation and is considered acceptable for garment production.

Expected Impact

The capstone element is expected to improve customer confidence through garment visualization [12], reduce measurement-related fitting errors [7], minimize production revisions and material waste [7], improve communication between customers and tailoring staff, and enhance overall customer satisfaction [1].

2.3.3 Data Dictionary

The data dictionary defines the primary database entities and their corresponding attributes.

Table 2. User Table

Field Name	Data Type	Description
user_id	INT (PK)	Unique identifier of the user
first_name	VARCHAR(50)	User's first name
last_name	VARCHAR(50)	User's last name
email	VARCHAR(50)	User email address
password	VARCHAR(50)	Encrypted password
role	ENUM	Role (Admin, Staff, Customer)
contact_number	VARCHAR(20)	User contact number
created_at	TIMESTAMP	Record creation date
updated_at	TIMESTAMP	Record modification date

Table 3. Customer Profile Table

Field Name	Data Type	Description
customer_id	INT (PK)	Unique customer identifier
user_id	INT (FK)	References User Table
gender	VARCHAR(20)	Customer gender
Address	TEXT	Customer address
profile_image	VARCHAR(50)	Profileimage path
created_at	TIMESTAMP	Record creation date

Table 4. Measurement Table

Field Name	Data Type	Description
measurement_id	INT(PK)	Measurement identifier
customer_id	INT (FK)	References Customer Table
chest	DECIMAL(5,2)	Chest measurement
waist	DECIMAL(5,2)	Waist measurement
hips	DECIMAL(5,2)	Hip measurement
shoulder	DECIMAL(5,2)	Shoulder measurement
sleeve_length	DECIMAL(5,2)	Sleeve length
inseam	DECIMAL(5,2)	Inseam measurement
date_recorded	DATE	Date record

Table 5. Appointment Table

Field Name	Data Type	Description
appoinment_id	INT(PK)	Appointment identifier
customer_id	INT(FK)	References Customer Table
appoinment_date	DATETIME	Scheduled appointment
purpose	VARCHAR(50)	Purpose of appointment

status	VARCHAR(50)	Pending, Approved, Completed
remarks	TEXT	Additional notes

Table 6. Service Order Table

Field Name	Data Type	Description
order_id	INT(PK)	Order identifier
customer_id	INT (FK)	References Customer Table
garment_type	VARCHAR(50)	Type of garment
garment_reference	VARCHAR(50)	Uploaded design image
order_description	TEXT	Order specifications
total_amount	DECIMAL(10,2)	Total order cost
order_status	VARCHAR(50)	Current order status
order_date	DATE	Date ordered

Table 7. Payment Table

Field Name	Data Type	Description
payment_id	INT (PK)	Payment identifier
order_id	INT (FK)	References Service Order Table
payment_method	VARCHAR(50)	Payment Type
amoount_paid	DECIMAL(10,2)	Payment amount
payment_date	DATETIME	Date of payment
payment_status	VARCHAR(20)	Paid, Partial,Pending

Table 8. Virtual Try-On Table

Field Name	Data Type	Description
tryon_id	INT(PK)	Virtual try on identifier
customer_id	INT (FK)	References Customer Table
uploaded_image	VARCHAR(50)	Customer uploaded image
garment_image	VARCHAR(50)	Selected garment image
generated_preview	VARCHAR(50)	Generated try on result
created_at	TIMESTAMP	Generation date

Table 9. Measurement Validation Table

Field Name	Data Type	Description
validation_id	INT (PK)	Validation Identifier
measurement_id	INT (FK)	References Measurement Table
variance_value	DECIMAL(5,2)	Computed variance
tolerance_value	DECIMAL(5,2)	Allowed tolerance
validation_result	VARCHAR(20)	Valid or Invalid
validation_date	DATETIME	Validation date

Table 10. Order Tracking Table

Field Name	Data Type	Description
tracking_id	INT (PK)	Tracking identifier
order_id	INT (FK)	References Service Order Table
status	VARCHAR(50)	Current production stage
updated_by	INT (FK)	Staff responsible
updated_time	DATETIME	Date and time updated
remarks	TEXT	Progress notes

Table 11. Notification Table

Field Name	Data Type	Description
notification_id	INT (PK)	Notification identifier
user_id	INT (FK)	Recipient user
title	VARCHAR(50)	Notification title
message	TEXT	Notification content
status	VARCHAR(20)	Read or Unread
created_at	TIMESTAMP	Notification date

2.3.4 Architecture Design

The proposed system follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

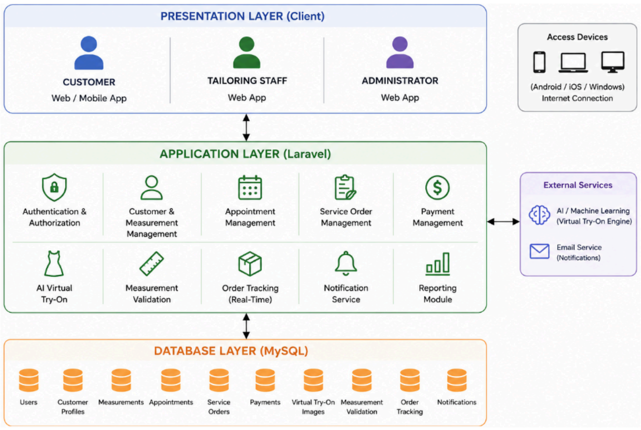

Figure 3. System Architecture Diagram

Presentation Layer

The Presentation Layer consists of the web and mobile interfaces used by Administrators, Tailoring Staff, and Customers. Users interact with the system through browsers or mobile devices.

Application Layer

The Application Layer contains the business logic implemented using Laravel Framework. This layer processes user requests, manages appointments, validates measurements, generates virtual try-on outputs, and handles order tracking operations.

Database Layer

The Database Layer utilizes MySQL Database Management System for storing customer information, measurements, appointments, service orders, payment records, and system-generated reports.

The architecture supports secure communication between system components while ensuring data integrity and efficient information processing.

2.3.5 Prototypes

The prototype phase presents the proposed user interface designs developed using Figma. These prototypes serve as visual representations of the final system and provide stakeholders with an overview of system navigation and functionality before implementation.

(image here)

Figure 4. Login Page Prototype

The Login Page allows users to authenticate their accounts by entering their registered email address and password. The interface also provides options for account registration and password recovery.

(image here)

Figure 5. Dashboard Prototype

The Dashboard serves as the main interface of the system. It displays summary information regarding appointments, service orders, payments, customer records, and order status updates.

(image here)

Figure 6. Appointment Management Prototype

This page allows customers to schedule appointments while enabling administrators and tailoring staff to manage fitting schedules and appointment requests.

(image here)

Figure 7. Virtual Try-On Prototype

The Virtual Try-On page allows customers to upload images and preview selected garment designs before production. The system generates visual representations based on available measurement and garment data.

(image here)

Figure 8. Real-Time Order Tracking Prototype

This page displays the current progress of customer orders, including pending, in production, fitting, completed, and released statuses. Customers can monitor garment progress without contacting tailoring personnel directly.

Figure 9. Payment Management Prototype

The Payment Management page records customer transactions and displays payment history, balances, and transaction details for monitoring and reporting purposes.

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